Business Case -Bayer Sr. HR People Data Analyst

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Methodology: The Scientific Approach to HR Data

Methology: A Scientific Approach to People Analyticis

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Phase 1: Exploratory Data Analysis (EDA)

Uncovering hidden patterns across **35+ variables.**

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Phase 2: Hypothesis-Driven Investigation

Moving beyond **"what happened"** to **"why it happened."**

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Phase 3: Segmentation

Analyzing data by **Job Role, Department, and Gender** to avoid **"average" bias.**

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Phase 4: Strategic Translation

Converting analytical insights into strategic actions with measurable business impact

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Technical Decisions & Data Architecture

Tools

Environment: Python (Pandas/Seaborn) for high-fidelity statistical plotting.

Data Cleaning: Managing categorical encodings and handling skewed distributions in MonthlyIncome.

Code Integrity: Used FutureWarning management to ensure a production-ready notebook.

Visualization Choice

→ **Boxplots** for salary and promotion to see the **"hidden" outliers.**

→ **Barplots with Error Bars** for satisfaction to ensure statistical significance.

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Compensation as a Retention Driver: A Segmented Reality

Hypothesis: Employees with salaries below the role's average are more likely to leave for better market opportunities.

✅ Validated (Entry-Level)

For **Sales Reps** and **Lab Techs**, talent loss occurs where the median salary is **15-20% lower** than that of retained employees.

❌ Invalidated (Leadership)

Managers who left were earning, on average, **+$1,000/month** compared to those who stayed, proving that salary is not the root cause of turnover at this level.

⚠️ Anomaly (HR)

100% of HR exits are concentrated at the **$2,500 salary base**, failing to capture the wide range of **$10,000+** observed among stable employees.

Money buys loyalty at the base, but it doesn't buy engagement at the top.

Implement a targeted market-adjustment for entry-level roles to plug the **"talent leak,"** while shifting the focus to culture and autonomy for senior leadership.

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Career Velocity: Timing the "Patience Threshold"

Hypothesis: Long-term stagnation (3+ years without promotion) is the main cause of career-related attrition.

The Findings (The Data)

The Paradox

Those who **stay** actually have a higher average time since their last promotion **(2.2 years)** than those who **leave (1.9 years).**

The Critical Window

Talent doesn't wait to be **"stagnated."** The decision to exit happens **just before the 2-year mark.**

> We are reacting too late. Our most ambitious talent leaves before they even hit the average promotion cycle of those who stay.

Do not wait for the 24-month mark. Standardize career pathing discussions at the **18-month milestone** to preempt the **"2-year itch."**

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Internal Equity: A Foundation of Meritocracy

Hypothesis: There is a systemic gender pay gap where men earn more than women in equivalent roles.

✅ Proven Equity

- In core technical and operational roles—such as **Research Scientist** (~$3,300) and **Laboratory Technician** (~$3,200)—average pay is virtually identical between genders.
- This confirms that the company's compensation structure is based strictly on role requirements rather than gender bias.

💪 Leadership Strength

- Women in **Manufacturing Director** roles ($7,400) and **Human Resources** ($4,500) show a higher average income than their male counterparts.
- This trend reflects successful long-term retention of female talent in high-seniority leadership positions.

> *The culture is built on Metric-Driven Equity. have a clean bill of health regarding pay discrimination, which is a massive asset for Employer Brand.*

Leverage these findings in **Employer Branding campaigns** to attract diverse high-level talent. Our data-proven equity is a **competitive advantage** in the talent market.

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The "Destination Office": Commute vs. Satisfaction

Hypothesis: Long commutes (15km+) act as an **"emotional tax,"** significantly reducing environment satisfaction.

The Findings (The Data)

Hypothesis Defied

Satisfaction levels remain **stable across all distance brackets.**

Office Quality

The office experience is **strong enough to neutralize** the fatigue of the commute.

The office is a destination worth the commute. Employees are willing to travel longer distances because the workplace consistently delivers a high-quality environment.

There is a clear geographic opportunity: the talent sourcing radius can be expanded without compromising engagement, as long as office culture and environmental standards are consistently maintained.

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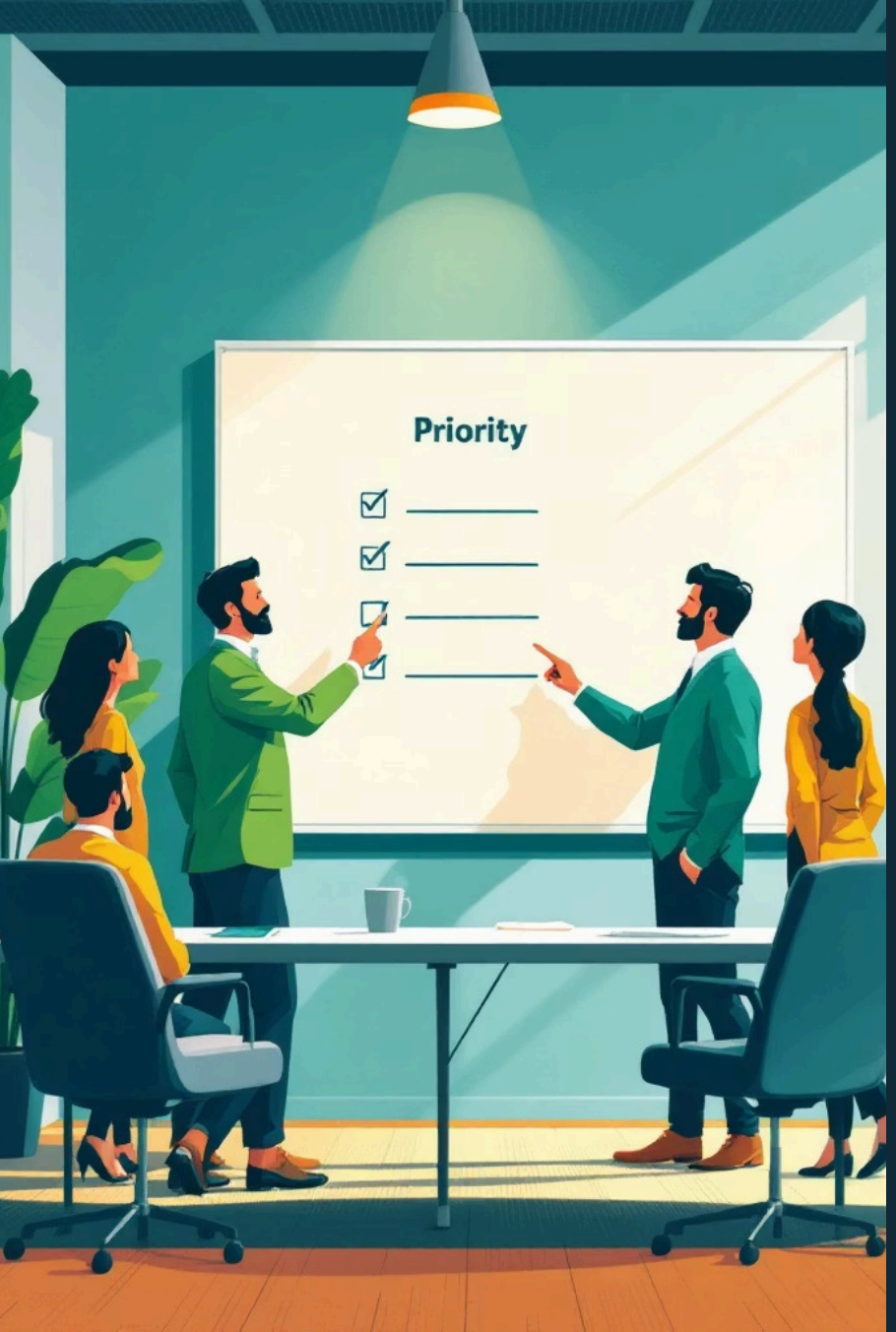

Priority Action Plan

Surgical Pay Adjustment

Target **Sales Reps and Lab Techs** to match market medians immediately.

The 18-Month Career Reset

Standardize career pathing talks **6 months before** the critical attrition window.

Performance-Linked L&D

Redesign training to specifically target the skills needed to achieve **Rating 4.**

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AI as a Strategic

Code Optimization

Accelerated the development workflow using **Gemini**, supporting decisions such as whether to download or work directly within Jupyter environments, and assisting in enhancing data visualizations with improved color schemes and clarity

Business Translation

Leveraged **Gamma** to design and structure the presentation, enabling a clear, visually compelling narrative of insights and recommendations.

Critique & Refinement

AI supported the translation of technical outputs (e.g., boxplot medians and statistical patterns)

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Data Quality & Future Scope

Limitations

Snapshot Data

The dataset is a **point-in-time snapshot;** we lack longitudinal data to see historical trends.

Missing Variables

We need **"Exit Interview" text data** to confirm the non-financial motives of departing Managers.

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